

Lainey Ward

PhD Student | Decarb-AI Centre | Innovate for Ireland iScholar
School of Civil Engineering, University College Dublin

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SUMMARY

PhD student at the University College Dublin Decarb-AI Centre (“Decarbonising Ireland: AI-powered pathways to climate resilience”) and an Innovate for Ireland iScholar, researching subseasonal-to-seasonal predictability of multi-hazard weather and climate events with a focus on machine-learning methods.

EDUCATION

2024 – 2028 **Doctor of Philosophy (PhD)**

Decarb-AI, School of Civil Engineering, University College Dublin

Supervisors: Dr. Fiachra O’Loughlin (Primary), Dr. Conor Sweeney

Thesis: Predictability of multi-hazard weather and climate events on subseasonal-to-seasonal timescales using machine learning

2023 – 2024 **MSc Atmosphere, Ocean and Climate – Distinction**

Department of Meteorology, University of Reading

Class: **Distinction**

Dissertation: Quantifying long-duration storage capacity and meteorological drivers of net-zero electricity systems

Relevance: atmospheric and ocean dynamics, physical meteorology, numerical modelling, and weather forecasting.

Award: UK Met Office MSc Dissertation Prize

2018 – 2022 **B.Sc. Physics**

University College Dublin

Class: **2:1**

Dissertation (A+): Computational analysis of water-harvesting materials; resulting in co-authorship of a Q1 peer-reviewed publication.

Award: UCD Entrance Scholar (2018)

WORK EXPERIENCE

Aeronautical Meteorological Officer, Met Éireann

Sep 2024 – Sep 2025

Provided operational aviation weather services in compliance with ICAO and EU regulations. Supported the Aviation Modernisation and Automation Project and contributed to data-rescue initiatives.

Tutor and Demonstrator, University College Dublin

Jan 2025 – Present

Delivered tutorials and practical sessions supporting undergraduate programming and mathematical skills, including problem-solving and assignment guidance.

PUBLICATIONS

Ward, L., Forero-Martinez, N., Cortés-Huerto, R., & Ballone, P. (2022). Water harvesting by thermoresponsive ionic liquids: A molecular dynamics study of the water absorption kinetics and the role of nanostructuring. *The Journal of Physical Chemistry B*, 127(24), 5494–5508. <https://doi.org/10.1021/acs.jpcb.3c01655>

SKILLS & COMPETENCIES

Programming	Proficient in Python for scientific computing, data analysis, numerical modelling, and machine learning; experience with MATLAB and R; experienced with Git and HPC computing clusters.
Machine Learning	Applied machine-learning methods in atmospheric and climate contexts, including clustering, feature-based analysis, and deep-learning models (e.g. CNNs, LSTMs, and Vision Transformers).
Data Analysis	Analysis of large meteorological and climate datasets, including reanalysis and forecast products; strong statistical analysis and visualisation skills.
Forecasting	Operational experience in weather forecasting and interpretation of observational and forecast products.
Communication	Clear scientific communication through research presentations, written reports, and undergraduate teaching.

ACHIEVEMENTS

Met Office MSc Dissertation Prize

Awarded for outstanding research for the MSc dissertation “Quantifying Long-Duration Storage Capacity and the Meteorological Drivers of Net-Zero Electricity Systems”.

UCD Entrance Scholar Award

Awarded for academic excellence on entry to University College Dublin (2018).

UCD Academic Excellence Award

Awarded for highest academic results (2017).

References available on request.